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Measurement of equivalent friction coefficient of tapered roller bearing utilising the theorem of kinetic energy

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Abstract

The friction performance of roller bearings serves as a critical performance indicator. Currently, friction torque is commonly utilised to evaluate the friction performance of roller bearings. The value of the friction torque is significantly influenced by the specific measurement conditions, e.g., the bearing size, applied load, etc. This work proposes a novel concept of equivalent friction coefficient for roller bearings, which is elaborately interpreted using a tapered roller bearing as a case study. A measurement method for determining the equivalent friction coefficient of roller bearings was developed according to the theorem of kinetic energy. A kinetic energy model of the measurement system, incorporating the centre shaft and bearing under measurement, was established. A measurement platform for the equivalent friction coefficient was constructed and utilised for a typical type of tapered roller bearing. The equivalent friction coefficients at different rotation speeds were determined utilising the platform. The investigation results possess significant theoretical and engineering value in enriching the evaluation methods for the friction performance of roller bearings.

Keywords

Tapered roller bearing, equivalent friction coefficient, friction torque, theorem of kinetic energy, friction performance

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Introduction

Rolling bearings are one of the fundamental components in the mechanical industry, extensively utilised in critical equipment including CNC machine tools, high-speed trains, and large shield tunnelling machines.[1] Friction consumption is the primary source of energy consumption in rolling bearings. Excessive friction in rolling bearings initiates a detrimental chain reaction: rising temperatures lead to lubrication degradation, which in turn causes vibrational disturbances and noise, ultimately compromising the operational stability of the total equipment.[2,3] Therefore, the accurate and comprehensive evaluation of the friction performance of rolling bearings has important theoretical and engineering significance.[4]

At present, the frictional torque is primarily utilised to evaluate the friction performance of rolling bearings, and the specific research methods can be classified into two categories: theoretical and experimental methods.[5] In terms of theoretical method, the Harris formula and SKF formula are the classical results for calculating the frictional torque of rolling bearings. The Harris formula divided the friction torque into two components, i.e., the component caused by the applied load and the component induced by the viscous drag of the lubricant.[6] The SKF formula further categorised the friction torque of rolling bearings into four distinct components: rolling friction torque, sliding friction torque, seal friction torque, and friction torque from drag loss.[7] Luc Houpert systematically formulated friction torque models for ball bearings and tapered roller bearings and performed a series of experiments. The sliding speed at the ball-race interface has been characterised and was responsible for torque contributions of the ball bearings: the torque resulting from the spatial curvature of the contact ellipse and the torque arising from spinning effects, while these torque contributions are absent in tapered rolling bearings.[8] Aihara enhanced the classic Harris formula by incorporating the rolling resistance and oil film thickness of elastohydrodynamic lubrication into the running friction torque of rolling bearings

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and achieved a high-precision prediction result.[9] Deng et al. conducted a systematic study on the theoretical analysis of the friction torque of rolling bearings utilising the SKF formula and dynamic analysis methods.[10,11] They established a theoretical model for calculating the friction torque of angular contact ball bearings based on the bearing dynamics and thermodynamics theory, and further analysed the effects of structural and operating parameters on the friction torque.[12] The experimental results showed that the theoretical model could accurately predict the friction torque of angular contact ball bearings throughout a spectrum of speeds, while the conventional model failed under the high-speed conditions with significant prediction errors. Theoretical study and experimental results demonstrated that as the axial load increases, the friction torque of the bearing also increases, and the axial load is basically proportional to the friction torque.[13] However, the proportionality coefficient varies at different speeds, mainly due to the increase in speed causing an increase in the thickness of the lubricant oil film between the rolling elements and the raceway, and hence reducing the proportionality coefficient. In addition, regarding the calculation of friction torque of ball bearings under solid lubrication/dry contact conditions, Ma et al. proposed a friction torque calculation method for ball bearings based on the rolling contact creep theory.[14] The contact issue between the rolling elements and the raceway was segmented into normal Hertz contact and tangential viscous subproblems, therefore revealing the rolling adhesion motion characteristics of the rolling elements relative to the inner and outer raceways. Zhang et al. proposed a friction torque calculation method considering the tilting effect of the tapered bearing rollers under dry friction conditions.[15] They demonstrated that an appropriate roller tilt angle can reduce the friction torque of the bearing, and the critical tilt angle is a quadratic function with regard to the friction coefficient and the clearance value of the cage pocket.

In the aforementioned theoretical model, several correction coefficients for working conditions must be introduced to accurately predict the friction torque of rolling bearings, and the values of these coefficients are derived from experimental research.[16] At present, there are three methods for measuring the friction torque of rolling bearings, i.e., the transmission method, the equilibrium method, and the energy method. Both the transmission method and the equilibrium method depend significantly on high-precision torque sensors.[17] Several business companies have produced equipment for measuring friction torque of rolling bearings, appropriate for diverse operational conditions.[18–20] In addition, some researchers have also conducted a series of studies on the development and experimental research of friction torque equipment. For instance, Ionut et al. performed numerical simulations and experimental investigations on the friction torque of radial ball bearings; and subsequently developed an experimental prototype for measuring the friction torque of ball bearings.[21] Cousseau et al. studied the effect of grease formulation on power loss in thrust bearings.[22–24] They claimed that the friction torque is influenced by the viscosity and properties of the grease base oil, the friction coefficient under full film conditions, and the interaction between the grease thickener and the base oil. Chen developed a test rig for measuring friction torque utilising the equilibrium method, which is capable of assessing friction torque under various working conditions, including clearance, speed, and load.[25] Their findings showed that as the bearing speed and radial load increase, the friction torque of the rolling bearing correspondingly increases, aligning with the calculations derived from the SKF formula. Tao et al. measured the friction torque of vehicle bearings under different radial forces and rotational speeds using the equilibrium method, and employed the least square method to fit the relationship between the friction torque and the bearing speed.[26] Their investigations indicated that the empirical formula obtained by polynomial function fitting has higher accuracy than the traditional Palmgren formula. Blake et al. designed a novel experimental setup that utilised springs to simultaneously apply loads to tapered roller bearings in both radial and axial directions.[18] The running friction torque under combined load conditions was measured using the transmission method. Goncalves et al. studied the influence of lubricant thickener composition on the friction torque of thrust ball bearings using the equilibrium method.[27] Their findings indicated that as the percentage of thickener components increases, the sliding friction coefficient between the bearing ball and the raceway decreases, resulting in a gradual decrease in the friction torque of the rolling bearing. In addition, with the increase of thickener content, the oil film thickness under high-speed conditions also increases, resulting in a reduction of the friction torque of the ball bearing.

However, academics have discovered that the sole use of friction torque to evaluate the friction performance of rolling bearings has certain drawbacks and deficiencies as a result of the ongoing development of pertinent research. Firstly, the friction torque is not an inherent characteristic of rolling bearings and is contingent upon specific conditions, e.g., the bearing size and rotation speed. The friction torque of larger rolling bearings is also relatively higher, which hinders the comparison of the friction performance of rolling bearings with different sizes.[28] Secondly, the equilibrium method is the primary technique employed to measure the friction torque of rolling bearings, and its measurement accuracy is significantly influenced by the accuracy of the torque sensor.

To address the aforementioned limitations, it is necessary to introduce a new measurement method to perform a more comprehensive and accurate analysis on the friction characteristics of rolling bearings. Currently, friction torque and friction coefficient are two indicators employed to evaluate the friction performance of bearings of journal bearings.[29–31] Motivated by this fact, this study proposes a novel concept of equivalent friction coefficient (EFC) for rolling bearings, which is applied to further characterise the friction performance of rolling bearings. A new measurement method for the EFCs of rolling bearings is implemented based on the kinetic energy theorem. Finally, a test platform is developed to ensure the accurate measurement of the EFC of tapered roller bearings.

Theoretical model

Definition of EFC for tapered roller bearings

The contact and friction conditions of rolling bearings are complex, involving the rolling elements in high-pair contact with both the inner and outer rings. It is challenging to evaluate the frictional characteristics of rolling bearings using the typical friction coefficients. This study proposes an equivalent sliding bearing model corresponding to rolling bearings, whereby the energy loss of the rolling bearing is equal to the friction loss of the equivalent sliding bearing. Based on this equivalence, the friction coefficient of the equivalent sliding bearing is defined as the EFC of the corresponding rolling bearing.

Figure 1 presents tapered roller bearing (TRB) and the corresponding equivalent sliding bearing models. This study focuses on the friction characteristics of the TRBs, and utilises the equivalent sliding bearings of TRBs as a case study to illustrate the definition method of ECF. Figure 1(a) shows a single row TRB with a contact angle of α and an average radius of R_m (defined as the average value of the inner and outer ring radii). Figure 1(b) shows an equivalent sliding bearing model, wherein the outer ring is fixed and the inner ring rotates. The contact angle and centre radius of the inner and outer rings of the equivalent sliding bearing are identical to the corresponding values of the actual rolling bearing. The inner and outer rings of the equivalent sliding bearing constitute a sliding pair at the mating surface, and its friction coefficient is defined as:

$$\mu = \frac{F_f}{F_n} \tag{1}$$

where F_f is the frictional force on the sliding pair; and F_n denotes the normal force on the sliding pair. The friction coefficient μ on the friction is defined as the EFC of the tapered roller bearing.

The mathematical relationship between EFC and the friction torque is further established. The relationship between tangential friction force and friction torque can be expressed as:

$$F_f = \frac{M}{R} \tag{2}$$

where M represents the friction torque of the TRB.

This study analyses the axial load situation of TRBs, and the relationship between the axial force (F_a) and the normal force (F_n) is expressed as:

$$F_n = \frac{F_a}{\sin\alpha} \tag{3}$$

where F_a is the axial load acted on the TRB.

The relationship between the EFC and the friction torque of the TRB can be derived according to combine Equations (1–3), which is expressed as

$$\mu = \frac{M \sin\alpha}{F_a R_m} \tag{4}$$

Measurement scheme

Measuring the tangential friction force F_a and normal force F_n on the sliding pair of equivalent sliding bearings is inherently challenging, making it impossible to directly obtain the EFC of rolling bearings according to equation (1). The EFC is obtained by using the relationship between the friction torque (M) and the EFC (μ) in equation (4). However, the measurement precision of the friction torque is highly dependent

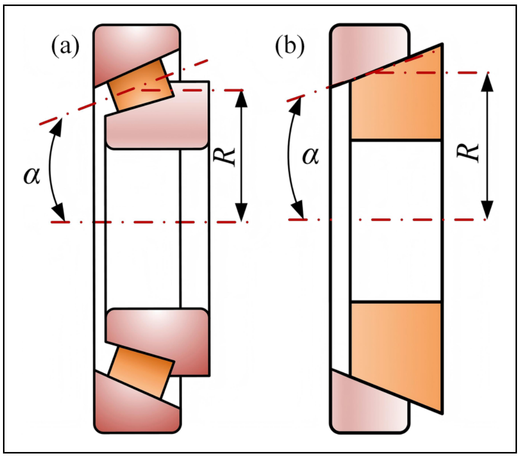

Figure 1. Tapered roller bearing (TRB) and its equivalent sliding bearing model: (a) components of TRB, including: 1-inner ring, 2-tapered roller, and 3-outer ring; and (b) components of the equivalent sliding bearing, including: 4-equivalent inner ring, and 5-equivalent outer ring.

on the accuracy of the torque sensor. To address this technological issue, we devised a way to utilise kinetic energy to determine the EFC of the rolling bearing, and the EFC can be accurately measured using high-precision speed sensors instead of the torque sensors.

Figure 2 shows the measurement principle for determining the EFC of the two TRBs based on the kinetic energy theorem under axial load conditions. Before measurement, the two bearings, designated A and B, were installed at the upper and lower ends of the main shaft. It is indeed that the EFCs of the two TRBs are different. A pressure plate was placed at the upper end to uniformly distribute the load on the tested bearings, and further apply different loads to the pressure plate using an electric cylinder. The inner ring of the TRB and the spindle rotated at the same speed vis a servo device, while the outer ring and the loading pressure plate remained stationary. The servo driving system accelerated the main shaft to the designated angular speed. Once the angular speed has stabilised, disconnect the clutch to allow the spindle to decelerate freely under the friction of the tested bearings. The angular speed of the main shaft corresponding to the deceleration time was simultaneously recorded using a high-precision speed sensor. The measured speed was transmitted to a computer through a high-speed reading head, and subsequently data processing yields the EFC value of the two TRBs.

Kinetic energy model

The kinetic energy model of the measuring system, incorporating the spindle and two tested TRBs, is established. Under the axial loading condition depicted in Figure 2, the angular speed of the outer ring of the bearing is zero. The rotation speed of the inner ring is the same as the main shaft (ω_c). The angular speed of the cage (ω_m) is the same as the roller revolution; and the angular speed of the rollers includes both the revolution speed (ω_m) around the inner ring and the spin speed (ω_z). The total kinetic energy of the measuring system is:

$$E_k = \frac{1}{2} J_c \omega_c^2 + J_n \omega_c^2 + J_b \omega_m^2 + n J_g \omega_m^2 + n J_z \omega_z^2 \tag{5}$$

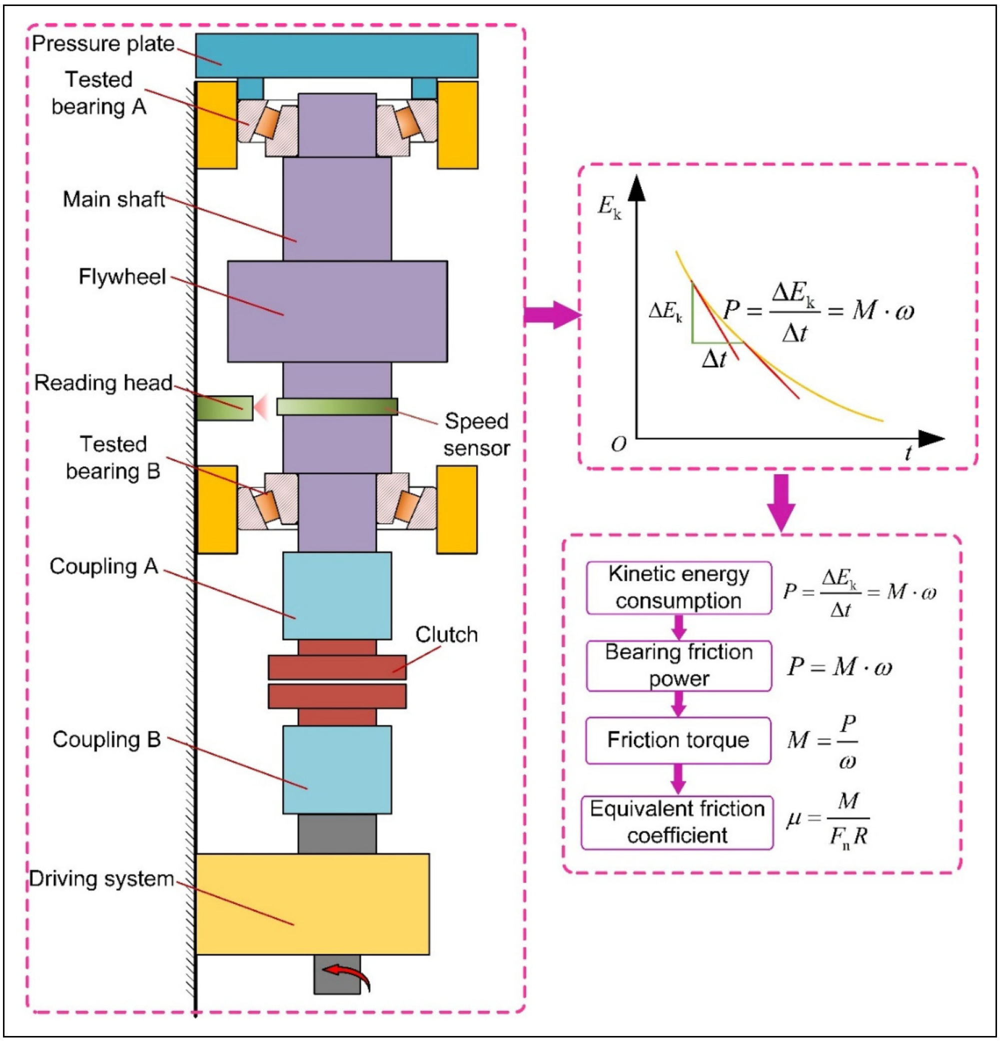

Figure 2. Measurement principle of the EFC for rolling bearing under axial load conditions.

where J_c, J_n, and J_b, are the moments of inertia of the shaft, inner ring and cage; J_g and J_z are the revolution and rotation moments of inertia for one bearing roller; n is the number of bearing rollers; ω_c is the angular speed of the main shaft; and ω_m and ω_z are the revolution and rotation speeds of the tapered rollers, respectively. For a given rolling bearing and main shaft system, the moment of inertia is a constant value, and the rotation speed varies with time. Equation (5) represents a time-dependent function, i.e., the kinetic energy-time model of the measurement system.

An accurate analysis of the speed relationship among the moving components of TRB is essential for a high-accuracy measuring system. The kinetic energy model presumably incorporates the following assumptions: (1) the rolling element undergoes pure rolling with the inner and outer rings, i.e., no slippage; (2) the geometric centre of the tested bearing coincides with its mass centre; and (3) the ambient temperature (22 °C) surrounding the tested bearings is maintained constant. Wang et al. pointed out that the contact between the tapered rollers and the raceway is characterised by linear contact. The contact between the tapered rollers and the raceway is nearly pure rolling contact, and the sliding effect between the roller and the raceway can be neglected.[32] The tested bearings were placed in a controlled-temperature laboratory, where minor temperature fluctuations have a negligible impact on the EFCs of the tested bearings. Under the condition of inner ring rotation, the speed relationship between the cage and the inner ring is derived as:[6]

$$\omega_m = k_1 \omega_c \tag{6}$$

where k_1 is the speed coefficient and calculated as:

$$k_1 = \frac{1}{2} - \frac{1}{2} \frac{\tan \beta}{\tan(\alpha - \beta)} \tag{7}$$

where β is the nominal value of the half cone angle of the tapered roller.

The angular speed of a tapered roller rotation is expressed as:[6]

$$\omega_z = -\frac{2k_1 k_2 D_m}{D_b} \omega_c \tag{8}$$

where the "-" indicates that the direction of angular speed of the tapered roller is opposite to the direction of the inner ring rotation. D_m is the average diameter of tapered roller bearings, and $D_m = 2R_m$. D_b is the nominal diameter of the tapered roller. k_2 is a coefficient and expressed as:

$$k_2 = \frac{1}{2} + \frac{1}{2} \frac{\tan \beta}{\tan(\alpha - \beta)} \tag{9}$$

By combining equations (5–6) and (8), the kinetic energy of the rotating system can be further expressed as:

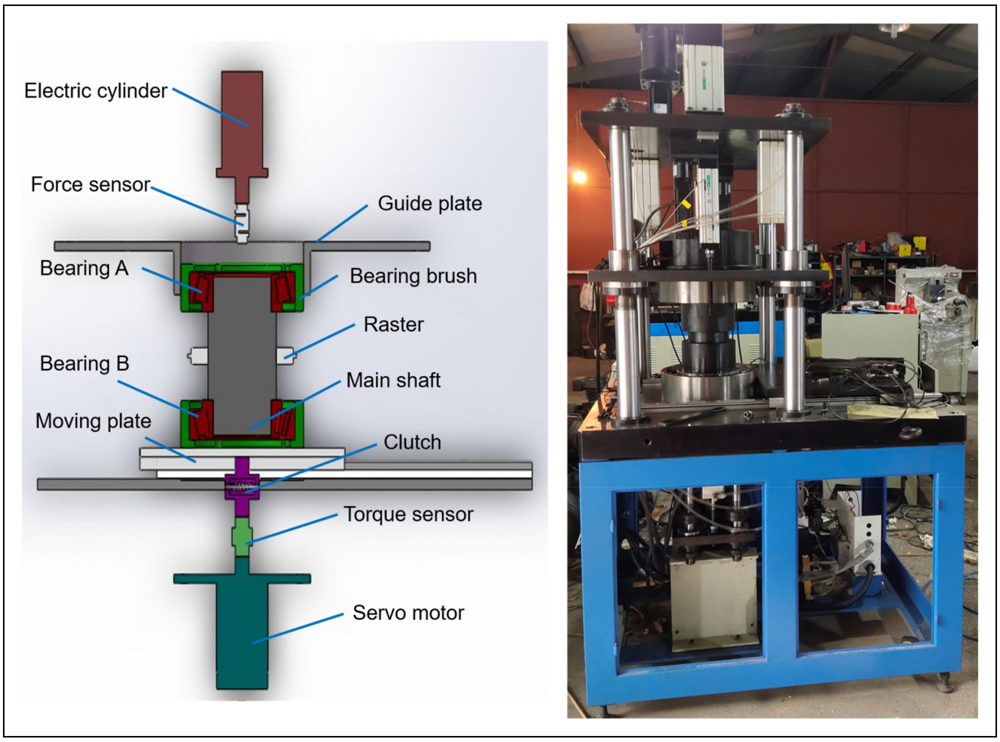

Figure 3. The measurement platform of equivalent friction coefficient for TRBs.

$$E_k = \frac{1}{2} J_e \omega_c^2 \tag{10}$$

where J_e is the equivalent moment of inertia of the spindle system, and its calculation formula is:

$$J_e = J_c + 2J_n + 2k_1^2 J_b + 2nk_1^2 J_g + 8n\left(\frac{D_m}{D_b}\right)^2 k_1^2 k_2^2 J_z \tag{11}$$

Calculation model of EFC

The calculation model of EFC is established as follows. The tested bearing A was located at the upper end of the main shaft, while the tested bearing B was located at the lower end. Once the clutch separated the main shaft from the driving device, the kinetic energy of the measuring system gradually decreased due to the friction torque exerted by the two tested bearings. Utilising the differential form of the kinetic energy theorem, the friction power of the tested bearings is calculated as

$$-\left(\mu_A \frac{F}{\sin\alpha} R_m + \mu_B \frac{F+G}{\sin\alpha} R_m\right)\omega_c = \frac{dE_k}{dt} \tag{12}$$

where F is the load applied by the electric cylinder to the pressure plate; G is the sum of the weights of the pressure plate, main shaft, and bearing A; μ_A and μ_B are the EFCs of the two tested bearings A and B, respectively. Equation (12) is further simplified as

$$\mu_A F R_m + \mu_B (F+G) R_m = -J_e \varepsilon_1 \sin\alpha \tag{13}$$

where ϵ_1 is the angular acceleration for the 1st measurement operation.

It is challenging to obtain the EFCs of the two bearings A and B from a single measurement operation. Therefore, a similar measurement operation was performed under identical operating conditions by swapping the positions of the two bearings of A and B, resulting in an equation similar to equation (13)

$$\mu_A (F+G) R_m + \mu_B F R_m = -J_e \varepsilon_2 \sin\alpha \tag{14}$$

where ϵ_2 is the angular acceleration of the 2nd measurement operation. Due to the difference in EFCs between two tested bearings A and B, there is a difference of $\epsilon_1 \neq \epsilon_2$ for the same ω_c. The EFCs of the two bearings A and B are obtained by combining equations (13–14), which is shown as

$$\begin{cases} \mu_A = \dfrac{F\varepsilon_1 - (F+G)\varepsilon_2}{2F+G}\left(\dfrac{J_e \sin\alpha}{GR_m}\right) \\ \mu_B = \dfrac{F\varepsilon_2 - (F+G)\varepsilon_1}{2F+G}\left(\dfrac{J_e \sin\alpha}{GR_m}\right) \end{cases} \tag{15}$$

According to the aforementioned process, we can simultaneously obtain the two EFCs of the tested bearings under different rotation speeds.

Results and discussions

Experimental setup

Utilising the established model of the EFCs of rolling bearings as discussed above, a measurement platform for measuring the EFCs of TRBs has been constructed. Figure 3 shows the schematic diagram and physical diagram of the measurement platform. The measurement platform consists of a machine body, a servo drive module, a main shaft, a loading module, and a software analysis system. A flywheel was mounted between the two tested bearings to further improve the rotational inertia of the spindle and extend the speed

Table 1. The moment of inertia of each part of spindle ($kg \cdot m^2$).

Name	Value	Name	Value
Shaft J_c	0.37193	Revolution roller J_g	0.00361
Inner ring J_n	0.04541	Spin roller J_z	0.00007
Cage J_b	0.00237	Total J_e	0.52388

deceleration time, which is beneficial to acquire more measurement data of tested bearing speeds and improve the measurement accuracy. The software analysis system was developed utilising LabVIEW.

The tested bearings (FL32234 tapered roller bearings) were supplied by the Wafangdian Bearing Group Co., Ltd, which was a single-row tapered roller bearing ($n = 20$). The moment of inertia for each part of the measuring system are shown in Table 1.

Furthermore, the contact angle of the TRB is $\alpha = 14.17°$, and the nominal value of the half cone angle of the tapered roller is $\beta = 2°$. Based on the two values, the speed coefficients are obtained as: $k_1 = 0.419$, and $k_2 = 0.581$. In addition, the gravity of the spindle system is $G = 1132.32$ N, the average radius R_m of the TRBs is 0.1226 m.

Measurement results and discussion

To reduce the influence of the surrounding environment on measurement results, strict environmental control was carried out throughout the measurement process. The measuring platform was placed in a temperature-controlled chamber with vibration isolation, maintaining an ambient temperature of 25 ± 0.5 °C. The influence of airflow in the measuring environment was also strictly controlled. Before measurement, the tested bearing must operate for about 30 min to achieve thermal equilibrium. The load applied by the electric cylinder during measurement was 500 N, and the rotation speed of the main shaft gradually decreased from 10 rad/s. The blue data points in Figure 4(a) show the deceleration curve of the tested bearing angular speed.

Previous results indicated that the polynomial fitting function was suitable for establishing the speed deceleration curve for rolling bearing.[26] Therefore, this study adopts the quadratic polynomial fitting to establish the mathematical relationship between the rotation speed and time. The obtained curve fit is shown by the red line in Figure 4(a). The relationship between the rotation speed and time, established by a quadratic polynomial, is as follows:

$$\omega_c = A_1 t^2 + B_1 t + C_1 \tag{16}$$

where A_1, B_1, and C_1 are the quadratic polynomial fitting coefficients for the 1st measuring operation. The correlation coefficient $R^2 = 0.997$ with the quadratic polynomial fitting, which proves that the fitting results are highly accurate. This result is significantly higher than the linear fitting ($R^2 = 0.863$) and approximates the cubic polynomial fitting ($R^2 = 0.997$).

The rotation acceleration during the 1st measurement is obtained according to the derivative of equation (16), i.e.,

$$\varepsilon_1 = 2A_1 t + B_1 \tag{17}$$

This study focuses on the variation of EFCs with angular speeds of TRB. Therefore, the variation of angular acceleration with angular speed of the tested bearing is obtained by combining equations (16–17):

$$\varepsilon_1 = -\sqrt{B_1^2 - 4A_1C_1 + 4A_1\omega_c} \tag{18}$$

The positions of bearings A and B were swapped, and the experiment was repeated under the identical working conditions. A similar method was employed to develop the relationship between the angular speed and acceleration, i.e.,

$$\omega_c = A_2 t^2 + B_2 t + C_2 \tag{19}$$

$$\varepsilon_2 = -\sqrt{B_2^2 - 4A_2C_2 + 4A_2\omega_c} \tag{20}$$

where A_2, B_2, and C_2 are the fitting coefficients for the 2nd measurement. The EFCs of the two TRBs can be obtained by substituting the fitting coefficients into equations (15–16), which are shown in Figure 4(b). Furthermore, the corresponding fitting coefficients are shown in Table 2.

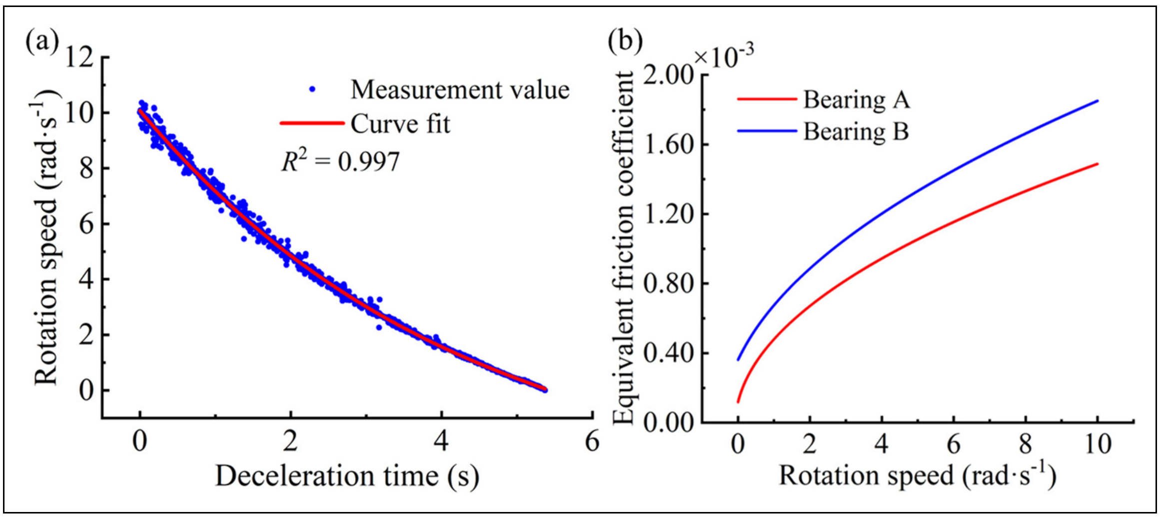

Figure 4. The variations of rotation speed and EFC for the two TRBs: (a) deceleration curve of main shaft vs. time; and (b) calculation results of EFCs for the two TRBs of A and B.

Table 2. The fitting coefficients for rotation speed of the two TRBs.

Item	Value	Item	Value
A_1	0.212	A_2	0.163
B_1	−2.954	B_2	−2.560
C_1	9.956	C_2	9.937

Table 3. The mean value and standard deviation of EFCs of both bearings A and B.

	1st measurement		2nd measurement		3rd measurement	
Tested bearing	Mean	S.D.	Mean	S.D.	Mean	S.D.
A	0.996×10^{-3}	0.415×10^{-3}	1.004×10^{-3}	0.392×10^{-3}	0.993×10^{-3}	0.401×10^{-3}
B	1.300×10^{-3}	0.421×10^{-3}	1.321×10^{-3}	0.417×10^{-3}	1.315×10^{-3}	0.415×10^{-3}

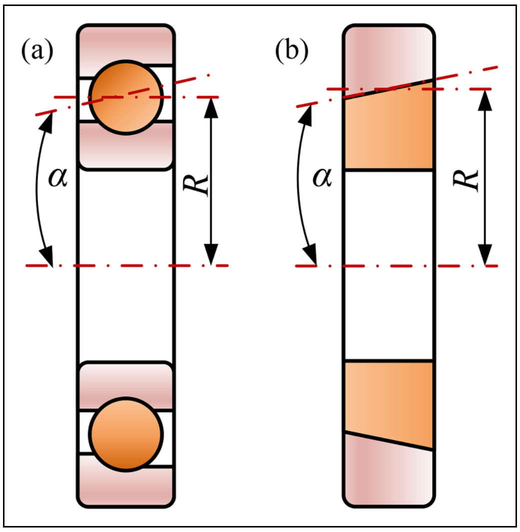

Figure 5. Angular contact ball bearing and its equivalent sliding bearing model.

The measurement results presented in Figure 4(b) indicate that as the rotation speed increases, the EFCs of both bearings also increase. However, the EFC of bearing A remains lower than that of bearing B. This discrepancy is expected, as variations in the machining and assembly processes of the inner ring, outer ring, and rolling elements inherently lead to unavoidable differences in the frictional characteristics of bearings A and B. To address this, the proposed method enables a reliable distinction to be made by swapping the positions of the tested bearings in the two measurement stages. Furthermore, the theoretical results of the SKF formula showed that the rotation speed of the TRB mainly affects the viscous friction torque component.[7] As the rotation speed increases, the viscous friction torque component increases, resulting in an increase in the friction torque and EFC of the rolling bearing. This result is consistent with the previous investigation results.[12,33]

To further verify the repeatability of the aforementioned results, two sets of measurement experiments were conducted under the identical measuring conditions, and the EFCs of the two bearings were measured separately. The average values of EFCs of the two bearings across the range of 0–10 rad/s are calculated, as shown in Table 3.

The data presented in Table 3 show that the mean EFCs of both bearings are approximately 1 ‰, which is consistent with previous empirical findings.[6] To further quantify the variability of the measurement results, the standard deviation (S.D.) has been introduced as a statistical metric.

$$S_\mu = \sqrt{\frac{\sum_{k=1}^{n} (\mu_k - \overline{\mu})^2}{n}} \tag{21}$$

where μ_k and $\overline{\mu}$ respectively represent the instantaneous value and the mean value of EFCs among the three groups for each tested bearing. The values of S.D. for TRBs A and B are respectively $(0.415–0.421)\times10^{-3}$ and $(0.401–0.415)\times10^{-3}$, indicating that the EFCs of rolling bearings have considerable repeatability.

In addition to the TRBs, the proposed method may also be applied to test other types and sizes of rolling bearings, including deep groove ball bearings, cylindrical roller bearings, and angular contact ball bearings, by utilising the various bearing adapters. A virtual sliding bearing model using the angular contact ball bearings is established to demonstrate this adaptability, as depicted in Figure 5. The calculation model for the EFCs of the two angular contact bearings is then derived from a similar kinetic energy model. The EFCs of the angular contact ball bearings are experimentally determined utilising the developed test apparatus.

Furthermore, to meet different rotational speed testing requirements, the desired measurement speed can be directly configured in the system. For instance, if measuring EFCs at rotational speeds below 20 rad/s is required, the system speed can simply be set to 20 rad/s. Similarly, the load can be adjusted by setting different load values on the servo electric cylinder within the system.

Conclusion

This study proposed a new quantitative index EFC to quantitatively evaluate the friction performance of rolling bearings. A measurement model for EFC was established, and a measurement platform was constructed to accurately measure EFC. Based on the aforementioned theoretical analysis and experimental results, some key conclusions are summarised as follows:

1. To comprehensively evaluate the friction characteristics of rolling bearings, this study proposed the

concept of EFC of rolling bearings. The friction coefficient of the equivalent sliding bearing was defined as the EFC of the actual rolling bearing. Based on this definition, a numerical relationship between the friction torque of a rolling bearing and the EFC was established.
2. A quadratic polynomial function was employed to establish the relationship between the deceleration relative to rotation time. The measurement results showed that the EFCs of TRBs were around 1 ‰ to 2 ‰, which is consistent with existing empirical results.
3. The EFC value of rolling bearings gradually increases with an increase in rotation speed, which is mainly due to the increase of rotation speed resulting in the increase of the viscous friction torque component of the friction torque. Different rolling bearings have different EFCs; therefore, EFC can be utilised to compare and evaluate the friction performance of rolling bearings.

Based on the measurement method, high-precision and repeatable measurement of the EFCs of rolling bearings was achieved, providing a novel approach for enriching the evaluation methods of friction characteristics of rolling bearings. Furthermore, this study focused solely on calculating and analysing the EFCs of tapered roller bearings (TRBs) under specific conditions, including particular load levels, bearing dimensions, and types. As such, it remains crucial to carry out further experimental investigations and analyses of EFCs across a broader range of rolling bearing types and loading scenarios. The findings of this research provide a novel and scientifically rigorous approach to characterising the frictional characteristics of rolling bearings. Future studies could extend this methodology to examine how EFCs vary under different design configurations and manufacturing processes, with the objective of determining the most favourable—i.e., the lowest—EFC values. By doing so, the EFC can evolve into a valuable guiding parameter for the design, production, and maintenance of rolling bearings, ultimately contributing to improved efficiency and performance in mechanical systems.

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Data availability

Data will be made available on request.

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